In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
data = {
    "Student_ID": range(1, 31),
    "Age": [18, 19, 20, 21, 19, 20, np.nan, 22, 21, 20,
            19, 18, 21, 20, 19, 23, 22, 20, 21, 19,
            18, 20, 21, 22, 19, 20, 21, 100, 19, 20],
    "Attendance_Percentage": [85, 90, 78, 88, 92, np.nan, 80, 75, 89, 91,
                              87, 85, 88, 90, 92, 30, 95, 89, 91, 87,
                              88, 90, 92, 94, 85, 86, 88, 90, 91, 92],
    "Study_Hours": [10, 12, 8, 15, 14, 11, 9, 7, 13, 12,
                    10, 9, 14, 11, 12, 2, 16, 13, 12, 11,
                    10, 12, 14, 15, 9, 10, 11, 50, 12, 13],
    "Exam_Score": [70, 75, 65, 80, 85, 72, 68, 60, 78, 82,
                   74, 70, 79, 76, np.nan, 55, 88, 80, 83, 74,
                   71, 77, 81, 85, 69, 72, 75, 95, 78, 80]
}

df = pd.DataFrame(data)
df
```

Out[2]:

	Student_ID	Age	Attendance_Percentage	Study_Hours	Exam_Score
0	1	18.0	85.0	10	70.0
1	2	19.0	90.0	12	75.0
2	3	20.0	78.0	8	65.0
3	4	21.0	88.0	15	80.0
4	5	19.0	92.0	14	85.0
5	6	20.0	NaN	11	72.0
6	7	NaN	80.0	9	68.0
7	8	22.0	75.0	7	60.0
8	9	21.0	89.0	13	78.0
9	10	20.0	91.0	12	82.0
10	11	19.0	87.0	10	74.0
11	12	18.0	85.0	9	70.0
12	13	21.0	88.0	14	79.0
13	14	20.0	90.0	11	76.0
14	15	19.0	92.0	12	NaN
15	16	23.0	30.0	2	55.0
16	17	22.0	95.0	16	88.0
17	18	20.0	89.0	13	80.0
18	19	21.0	91.0	12	83.0
19	20	19.0	87.0	11	74.0
20	21	18.0	88.0	10	71.0
21	22	20.0	90.0	12	77.0
22	23	21.0	92.0	14	81.0
23	24	22.0	94.0	15	85.0
24	25	19.0	85.0	9	69.0
25	26	20.0	86.0	10	72.0
26	27	21.0	88.0	11	75.0
27	28	100.0	90.0	50	95.0
28	29	19.0	91.0	12	78.0
29	30	20.0	92.0	13	80.0

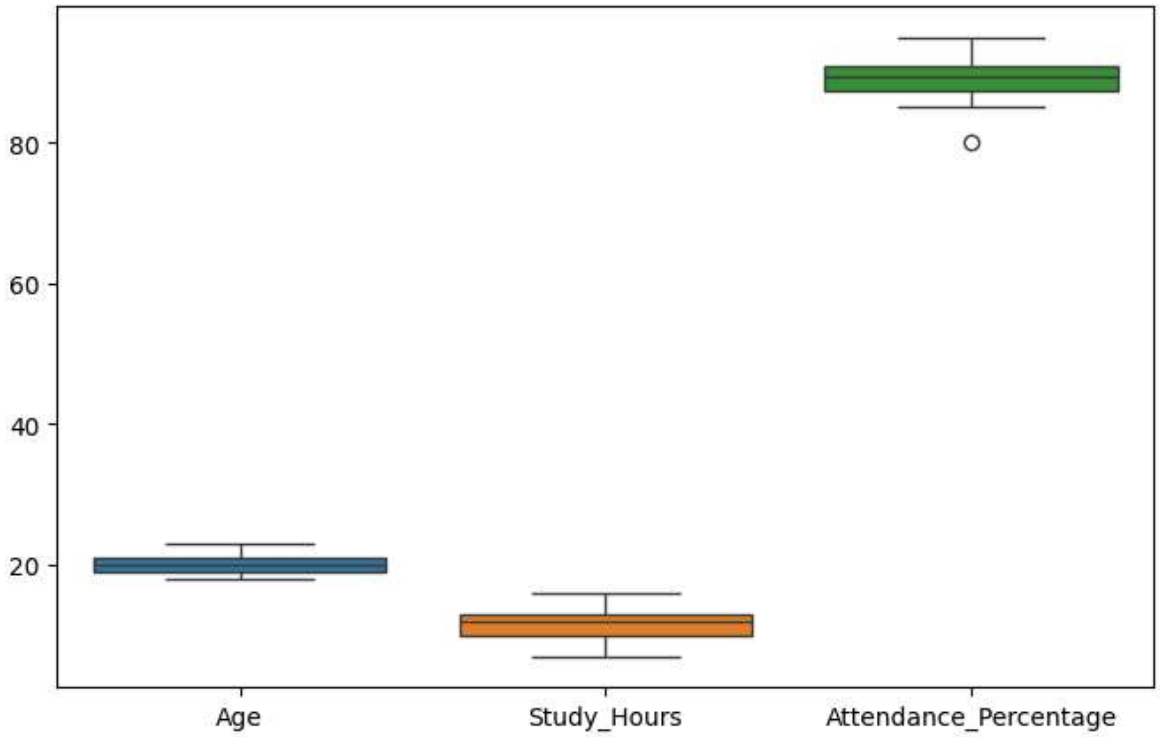
In [3]:
```
df.isnull().sum()
```

Out[3]:
```
Student_ID               0
Age                      1
Attendance_Percentage    1
Study_Hours              0
Exam_Score               1
dtype: int64
```

In [4]:
```
def detect_outliers_iqr(series):
    Q1 = series.quantile(0.25)
    Q3 = series.quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    return lower, upper
```

In [5]:
```
for col in ["Age", "Attendance_Percentage", "Study_Hours", "Exam_Score"]:
    lower, upper = detect_outliers_iqr(df[col])
    df.loc[(df[col] < lower) | (df[col] > upper), col] = np.nan
```

In [6]:
```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df[["Age", "Study_Hours", "Attendance_Percentage"]])
plt.show()
```



In [7]:
```
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Attendance_Percentage"] = df["Attendance_Percentage"].fillna(
    df["Attendance_Percentage"].mean()
)
df["Exam_Score"] = df["Exam_Score"].fillna(df["Exam_Score"].mean())
df["Study_Hours"] = df["Study_Hours"].fillna(df["Study_Hours"].mean())
```

In [8]:
```
df["Attendance"] = df["Attendance_Percentage"] / 100
```

In [9]:
```
df["Log_Study_Hours"] = np.log(df["Study_Hours"])
```

In [10]:
```
df = df.round(2)
df
```

Out[10]:

	Student_ID	Age	Attendance_Percentage	Study_Hours	Exam_Score	Attendance	L
0	1	18.0	85.00	10.00	70.00	0.85	
1	2	19.0	90.00	12.00	75.00	0.90	
2	3	20.0	89.04	8.00	65.00	0.89	
3	4	21.0	88.00	15.00	80.00	0.88	
4	5	19.0	92.00	14.00	85.00	0.92	
5	6	20.0	89.04	11.00	72.00	0.89	
6	7	20.0	80.00	9.00	68.00	0.80	
7	8	22.0	89.04	7.00	60.00	0.89	
8	9	21.0	89.00	13.00	78.00	0.89	
9	10	20.0	91.00	12.00	82.00	0.91	
10	11	19.0	87.00	10.00	74.00	0.87	
11	12	18.0	85.00	9.00	70.00	0.85	
12	13	21.0	88.00	14.00	79.00	0.88	
13	14	20.0	90.00	11.00	76.00	0.90	
14	15	19.0	92.00	12.00	75.81	0.92	
15	16	23.0	89.04	11.61	75.81	0.89	
16	17	22.0	95.00	16.00	88.00	0.95	
17	18	20.0	89.00	13.00	80.00	0.89	
18	19	21.0	91.00	12.00	83.00	0.91	
19	20	19.0	87.00	11.00	74.00	0.87	
20	21	18.0	88.00	10.00	71.00	0.88	
21	22	20.0	90.00	12.00	77.00	0.90	
22	23	21.0	92.00	14.00	81.00	0.92	
23	24	22.0	94.00	15.00	85.00	0.94	
24	25	19.0	85.00	9.00	69.00	0.85	
25	26	20.0	86.00	10.00	72.00	0.86	
26	27	21.0	88.00	11.00	75.00	0.88	
27	28	20.0	90.00	11.61	75.81	0.90	
28	29	19.0	91.00	12.00	78.00	0.91	
29	30	20.0	92.00	13.00	80.00	0.92	

In [ ]:

```

```